



SMOE



Ivar Aasen Field Development Project – pdQ

Review does not constitute acceptance of design details, calculations, test methods or materials developed or selected by vendor, nor does it relieve vendor from full compliance with contractual or other obligations.

- ☐ 1. Accepted
- ☐ 2. Accepted with comments incorporated
- ☐ 3. NOT accepted, revise and resubmit
- ☒ 4. Issued for Information Only
- ☐ 5. Interface Information as clouded (is accepted and frozen)

Name: Viren Modi

Date: 16/05/2015

01E	04/05/2015	Issued for Review	RH	OHP	ML
Rev.	Date	Reason for Issue	Prepared by	Reviewed by	Approved by
		VENTURI METERS CALIBRATION REPORT (GAS FLOW) - TAG no. 27FE 1101			
Package Title: EICT Package					
Area Code: M310 M410	System Code: 21, 27, 29, 44	Project Tag Numbers: 27FE1101			No. of Pages: 6
Project No.: C-0011		Supplier Document Number: Gas calibration cert.			Rev. 0
Purchase Order No: S09011		Project Document Number: DN02-S09011-I-RA-6568			Proj Rev. 01E



Calibration Certificate

Task No.: 115-23368.001
Certificate No.: 9.8-14229
Certificate date: 04-05-2015
Page: 1 of 3
Enclosure: 1 Gas mixture, 2 Calculation of the venturi tube.

CUSTOMER EQUIPMENT DATA:

Meter type: Venturi-meter
Size: 16INCH/ANSI600/DN400
No. / Year: 2011135-3/2015
TAG. No.: 27FE1101
Manufacturer: EMCO Controls

CUSTOMER:

Emco Controls A/S
Høgevej 6
3400 Hillerød
Att.: René Hansen

Order No.: -

TEST CONDITIONS:

Test method/medium: Volume flow of natural gas with a static pressure of 33,5 bar g.
Barometer reading: -
Ambient temperature: 20 ± 3°C

SCOPE OF TEST:

Calibration at the flow 960, 3000, 4800, 8400 and 12000 m³/h.
(3 repetitions).

FOUNDATION:

FORCE No. 60.10.4.2 Calibration with 24 INCH HPCL no. C03-002

REMARKS:

Technical evaluation: No Remarks.

TRACEABILITY:

Results: See page 2.
Test equipment: FORCE no. C03-001
See page 3.

DATE OF CALIBRATION:

25-03-2015

Responsible:


Jesper Busk
Signatory

Carried out by:


Thomas Boy Jacobsen

FORCE Technology, Division for Calibration and Measurement, Navervej 1 6600 Vejen phone: 76961600
Danish National Reference laboratory for volume gas measurement.

Doc. no: DN02-S09011-I-RA-6568

Doc. title: VENTURI METERS CALIBRATION

Rev. 01E

The *General Conditions* on the reverse page are an integrated part of our services.
Extracts from the certificate may only be reproduced with a written permission from FORCE Technology
The test results only include the tested objects.

Page 2 of 6

Results

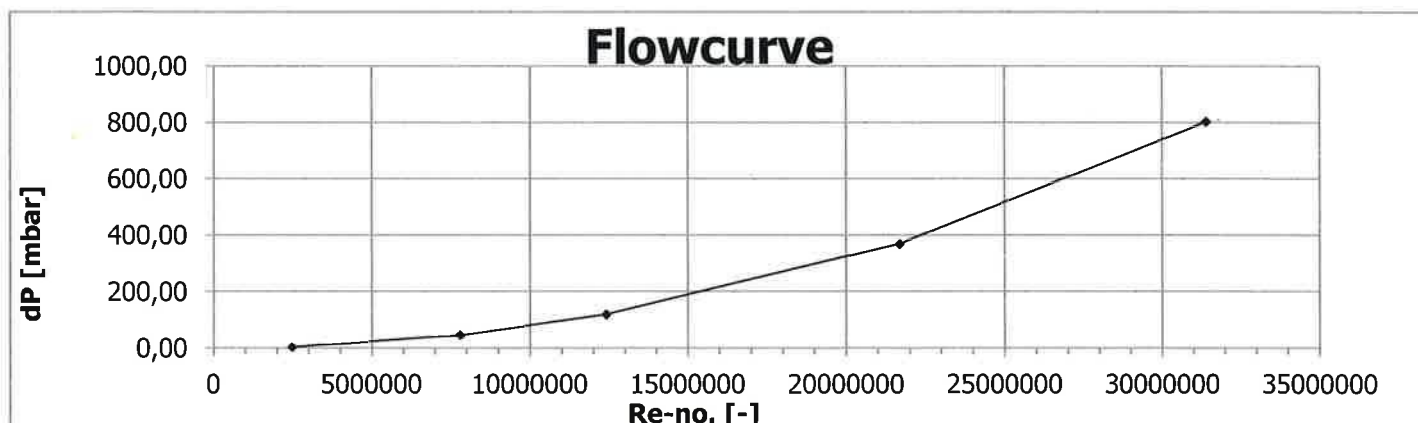
Volume meters (gas)

Meterdata:

Meter no. : 2011135-3
Metertype : Venturi-meter
Size : 16INCH/ANSI600/DN400

Calibration results 33,5 barg

Real flow	Calibration Uncertainty	Re no.	Measured dP	Reading uncertainty	Abs. Pressure	Temp.
[m³/h]	±Ucmc (%)	-	dP [mbar]	dP [mbar]	[bara]	[°C]
951,50	0,20	2480567	4,25	0,24	34,633	17,29
952,02		2481794	4,26	0,24	34,635	17,31
952,02		2481611	4,26	0,24	34,637	17,33
2999,28	0,19	7807008	47,14	0,41	34,570	17,24
2997,83		7801932	47,03	0,41	34,575	17,28
2998,34		7802717	47,07	0,41	34,578	17,30
4765,04	0,18	12401674	119,13	0,41	34,560	17,21
4766,34		12404022	119,09	0,41	34,562	17,23
4766,39		12403703	119,08	0,41	34,565	17,25
8335,29	0,18	21685732	370,84	0,41	34,486	16,95
8335,40		21683743	370,88	0,41	34,491	16,98
8335,66		21681943	370,87	0,41	34,495	17,01
12084,57	0,18	31390154	802,92	0,44	34,409	16,84
12083,17		31391392	802,83	0,44	34,405	16,80
12084,57		31398763	802,64	0,44	34,404	16,78



LABORATORY CONTROL EQUIPMENT

Equipment marked with x have been used during the calibration.

Working- Standards	FORCE No.	Traceability	Calibration range	Certificate No.
<u>Facilities: FORCE No. C03-002.</u>				
x Turbine meter 10 - 400 m ³ /h	A000590	PTB"	3 bar - 40 bar	9.8-13967
x Turbine meter 120 - 1000 m ³ /h	A000589	PTB"	3 bar - 40 bar	9.8-13966
x Turbine meter 250 - 4000 m ³ /h	A000584	PTB"	3 bar - 40 bar	9.8-13962
x Turbine meter 250 - 4000 m ³ /h	A000585	PTB"	3 bar - 40 bar	9.8-13963
Turbine meter 250 - 4000 m ³ /h	A000586	PTB"	3 bar - 40 bar	9.8-13964
Turbine meter 250 - 4000 m ³ /h	A000587	PTB"	3 bar - 40 bar	9.8-13965

Other equipment

x Temperature meters	A80037 & A80049
x Pressure meters	A20055 & A20059
x Pressure transmitter 100mbar no.	A80243
x Pressure transmitter 500mbar no.	A80241
X Pressure transmitter 1000mbar no.	A80242
x Digital multimeter	A00078
Digital multimeter	A00079

Laboratory accuracy:

Expanded uncertainty (for K=2) corresponds to a coverage probability of approx 95%

The standard uncertainty of measurement is in accordance with EA-4/02:

The accuracy for every calibration points is listed on page 2.

* PTB, via FORCE Technology's Travelstandards.
Doc. no: DN02-S09011-I-RA-6568

Doc. title: VENTURI METERS CALIBRATION
REPORT (GAS FLOW) - TAG no. 27FE 1101

Rev. 01E
Page 4 of 6

Task No.: : 115-23368.001
Certificate No.: : 9.8-14229
Certificate date: : 04-05-2015

Gasmixture during the calibration

The determination of the gasmixture is not covered by the DANAK accreditation no. 9

Description	Name	xj	S1
CH4	Methane	mol%	88,8097
C2H6	Etan	mol%	6,0639
C3H8	Propan	mol%	2,4430
n-C4H10	n-Butan	mol%	0,5929
i-C4H10	i-Butan	mol%	0,4027
n-C5H12	n-Pentan	mol%	0,0920
i-C5H12	i-Pentan	mol%	0,1400
C6+	Hexan +	mol%	0,0699
CO2	Carbon dioxide	mol%	1,0182
N2	Nitrogen	mol%	0,3677
		mol%	100,0

Task No.: 115-23368.001

Certificate No.: 9.8-14229

Certificate date: 04-05-2015

Enclosure 2

Calculation of the venturi tube

Client: Emco controls

$$q_V = C \cdot \varepsilon \cdot \frac{1}{\sqrt{1 - \beta^4}} \cdot \frac{\pi}{4} \cdot d^2 \cdot \sqrt{\frac{2 \cdot \Delta p}{\rho}}$$

$$\varepsilon = \sqrt{\left(\frac{\kappa \cdot \tau^{2/\kappa}}{\kappa - 1} \right) \cdot \left(\frac{1 - \beta^4}{1 - \beta^4 \cdot \tau^{2/\kappa}} \right) \cdot \left(\frac{1 - \tau^{(\kappa-1)/\kappa}}{1 - \tau} \right)}$$

q_V	=	Volume flow rate	[m ³ /s]
C	=	Discharge coefficient	[-]
ε	=	Expansion factor	[-]
β	=	Diameter Ratio d / D	[-]
d	=	Diameter throat	[m]
Δp	=	Differential pressure	[Pa]
ρ	=	Density	[kg/m ³]
κ	=	Isentropic exponent (C_p/C_v)	[-]
τ	=	Pressure Conditions (P_2/P_1)	[-]

Flow [m ³ /h]	Flow [m ³ /s]	d [mm]	D [mm]	β [-]	Δp [mbar]	Temp. [°C]	Pressure [mbar]	Pressure [bar]	κ [-]	τ [-]	ε [-]	ρ [kg/m ³]	C
951,5013	0,26431	232,693	381	0,61074278	4,661	17,29	34632,6	34,633	1,42	1,000	1,000	29,1261	1,019
952,0207	0,26445	232,693	381	0,61074278	4,667	17,31	34634,8	34,635	1,42	1,000	1,000	29,1257	1,019
952,0208	0,26445	232,693	381	0,61074278	4,665	17,33	34637,5	34,637	1,42	1,000	1,000	29,1256	1,020
2999,2764	0,83313	232,693	381	0,61074278	47,552	17,24	34570,5	34,570	1,42	0,999	0,999	29,0764	1,006
2997,8257	0,83273	232,693	381	0,61074278	47,440	17,28	34575,0	34,575	1,42	0,999	0,999	29,0749	1,007
2998,3444	0,83287	232,693	381	0,61074278	47,482	17,30	34577,9	34,578	1,42	0,999	0,999	29,0747	1,006
4765,0426	1,32362	232,693	381	0,61074278	119,533	17,21	34559,7	34,560	1,42	0,997	0,998	29,0704	1,009
4766,3395	1,32398	232,693	381	0,61074278	119,500	17,23	34561,8	34,562	1,42	0,997	0,998	29,0696	1,010
4766,3915	1,32400	232,693	381	0,61074278	119,490	17,25	34565,0	34,565	1,42	0,997	0,998	29,0700	1,010
8335,2944	2,31536	232,693	381	0,61074278	371,251	16,95	34486,4	34,486	1,42	0,989	0,993	29,0382	1,006
8335,3997	2,31539	232,693	381	0,61074278	371,293	16,98	34490,8	34,491	1,42	0,989	0,993	29,0378	1,006
8335,6611	2,31546	232,693	381	0,61074278	371,287	17,01	34494,5	34,495	1,42	0,989	0,993	29,0370	1,006
12084,5707	3,35683	232,693	381	0,61074278	803,356	16,84	34409,4	34,409	1,42	0,977	0,985	28,9826	0,999
12083,1671	3,35644	232,693	381	0,61074278	803,265	16,80	34404,8	34,405	1,42	0,977	0,985	28,9838	0,999
12084,5674	3,35682	232,693	381	0,61074278	803,073	16,78	34404,4	34,404	1,35	0,977	0,984	28,9858	1,000